ASSIGNMENT REPORT-10

Submitted by :

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Under the guidance of :

Dr. Rajalakshmi

For The Subject

Full Stack Programming



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING SHIV NADAR UNIVERSITY KALAVAKAM

1. Introduction

Student Grievance and Feedback Management System is a full-stack application developed using React JS, Node.js and MySQL. It provides a common platform for students to submit complaints and feedback, faculty members to handle assigned grievances, and administrators to monitor their resolution.

1.1 Objective

- Students log in with college email credentials, submit grievances with optional supporting documents, choose confidentiality and track responses.
- Faculty members view assigned complaints, add remarks or resolution notes and update the grievance status.
- Administrators assign officers, monitor pending complaints and view grievance statistics and feedback summaries.
- The back end validates input, enforces role and ownership permissions, stores salted password hashes and maintains a history of changes.

The database college_grievance_db contains the five required tables: users, grievances, grievance_responses, grievance_history and feedback. Primary keys, foreign keys and check constraints maintain valid relationships and ratings.

1.2 How to Run

1. Start MySQL. In app/.env, enter the MySQL credentials and choose DEMO_PASSWORD for the sample college accounts.
2. From the app folder, run npm install, then npm run db:setup.
3. Run npm run build, followed by npm start.
4. Open http://localhost:3000 and sign in with a sample account listed in README.md.

2. Tags Used

Tag / Function / API	Purpose
<header>, <main>, <nav>, <footer>	Organize the page header, dashboard navigation, main content and footer.
<form>, <label>, <input>	Provide accessible login, date, subject, file and confidentiality fields.
<select>, <option>, <textarea>	Select categories, ratings, officers and statuses; enter descriptions and responses.
<button>	Submit forms, apply filters, view complaints and sign out.
<table>, <thead>, <tbody>, <tr>, <th>, <td>	Display grievances, change history, statistics and feedback summaries.
createRoot, useState, useEffect	Render React components, manage form/dashboard state and load API data.
fetch()	Send login, grievance, response and feedback requests to the Node.js server.
express.json(), app.get(), app.post(), app.put()	Read JSON input and implement the required REST APIs.
mysql2.execute(), transactions	Use parameterized SQL and save grievance changes together with their history.
crypto.scrypt(), randomBytes(), timingSafeEqual()	Hash passwords with separate salts and verify passwords securely.
PRIMARY KEY, FOREIGN KEY, CHECK	Identify records, preserve relationships and restrict valid ratings and confidentiality values.

2.1 REST APIs Used

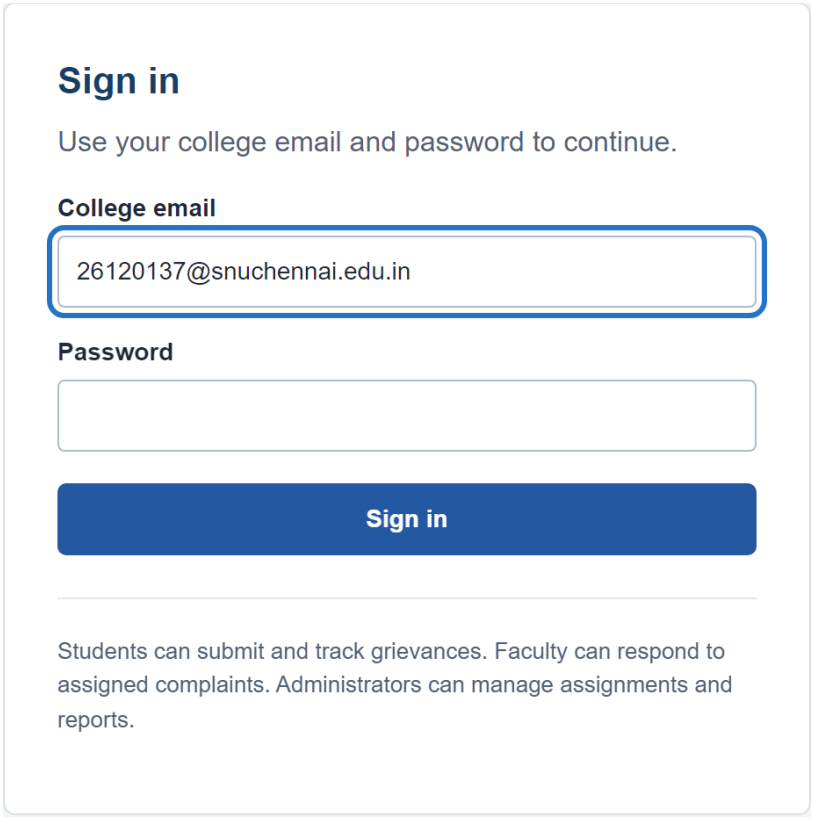
Method	Endpoint	Purpose
POST	/auth/login	Authenticate college users.
POST	/grievances	Submit a student grievance.
GET	/grievances/my	List the logged-in student's complaints.
GET	/grievances/{id}	View authorized complaint details, responses and history.
GET	/grievances/assigned	List a faculty member's assigned complaints.
PUT	/grievances/{id}/assign	Assign a complaint to a faculty member or officer.
PUT	/grievances/{id}/status	Update the complaint status.
POST	/grievances/{id}/responses	Add a response or resolution note.
POST	/feedback	Submit a service rating and comments.
GET	/feedback/summary	View administrator feedback summaries.
GET	/admin/reports	Generate complaint statistics.

Supporting routes handle the current login session, logout, the administrator complaint list, the faculty selector and authorized document downloads. Category, status and submission-date filters are applied in the back end.

3. Output Screenshots

The following browser captures show the React interface with sample API responses. Live database verification follows the final MySQL credential setup.

3.1 Login Page



College email and password are used to open the appropriate dashboard.

3.2 Grievance Submission

Submit a grievance

Describe the issue so the college can direct it to the right person.

Category

Infrastructure

Date

25-10-2026

Subject

Laboratory computers need maintenance

Description

Two computers in the laboratory freeze during programming sessions. Please arrange maintenance so students can complete their practical work.

Supporting document (optional)

Choose file laboratory-note.pdf

PDF, PNG or JPEG, up to 2 MB.

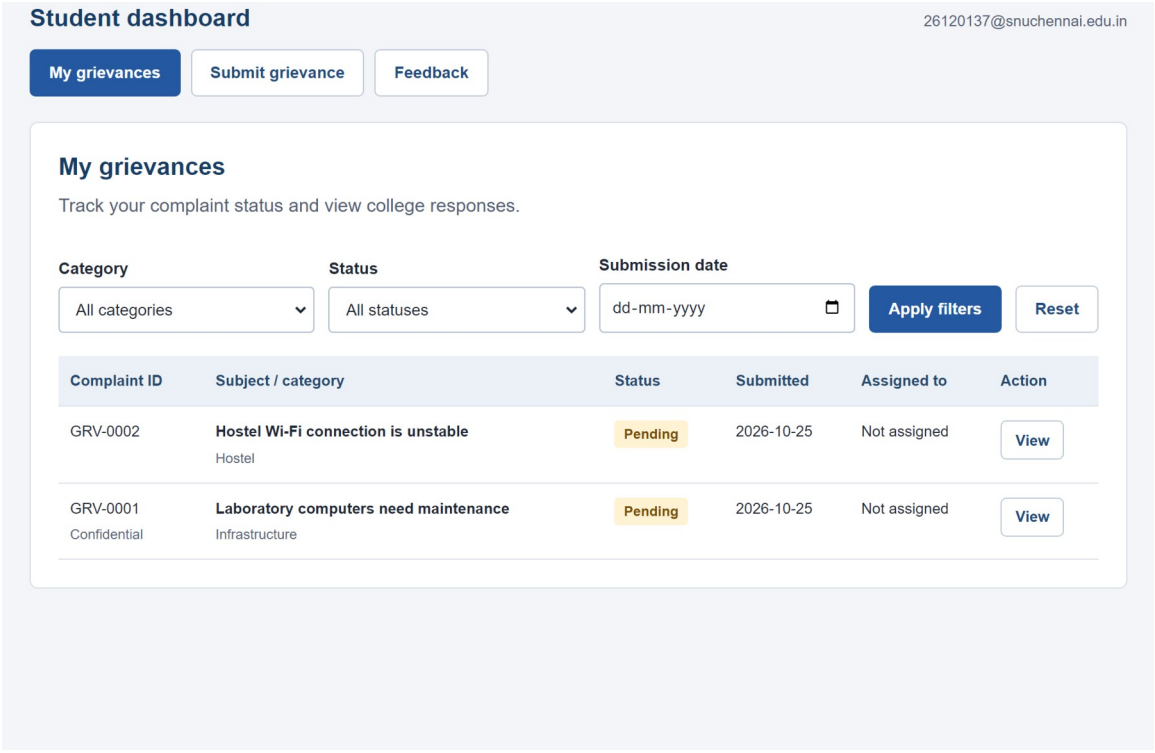
Mark this grievance as confidential

Confidential details are visible only to you, the assigned faculty member and the administrator.

Submit grievance

The student enters the category, subject, description and date, attaches a supporting document and selects confidentiality.

3.3 Student Grievance List



Students can view unique complaint IDs, status and assignments, and filter by category, status and submission date.

3.4 Administrator Assignment

GRV-0001: Laboratory computers need maintenance

Close details

Infrastructure · Submitted 25 Oct 2026, 07:15 pm

Pending | Confidential | Student: **Kamal Sundar SK** | Assigned to: **Dr. Rajalakshmi** | Grievance date: 2026-10-25

Two computers in the laboratory freeze during programming sessions. Please arrange maintenance so students can complete their practical work.

Download supporting document: laboratory-note.pdf

Assign faculty / officer

Dr. Rajalakshmi

Assign grievance

Update status

Status

Pending

Remarks (optional)

Save status

Add a response

Response / resolution note

Send response

Responses

No responses yet.

Grievance history

Date	Updated by	Status change	Remarks
25 Oct 2026, 07:15 pm	Kamal Sundar SK	Pending	Grievance submitted.
25 Oct 2026, 07:15 pm	College Administrator	Pending	Assignment: Unassigned to Dr. Rajalakshmi.

The administrator assigns the complaint to a faculty member. The assignment is included in the history.

3.5 Faculty Response and Status Update

GRV-0001: Laboratory computers need maintenance

Close details

Infrastructure · Submitted 25 Oct 2026, 07:15 pm

Resolved | Confidential | Student: **Kamal Sundar SK** | Assigned to: **Dr. Rajalakshmi** | Grievance date: 2026-10-25

Two computers in the laboratory freeze during programming sessions. Please arrange maintenance so students can complete their practical work.

Download supporting document: laboratory-note.pdf

Update status

Status

Resolved

Remarks (optional)

Save status

Add a response

Response / resolution note

Send response

Responses

Dr. Rajalakshmi 25 Oct 2026, 07:15 pm

The laboratory computers have been repaired and tested. Students can now use them for practical sessions.

Grievance history

Date	Updated by	Status change	Remarks
25 Oct 2026, 07:15 pm	Kamal Sundar SK	Pending	Grievance submitted.
25 Oct 2026, 07:15 pm	College Administrator	Pending	Assignment: Unassigned to Dr. Rajalakshmi.
25 Oct 2026, 07:15 pm	Dr. Rajalakshmi	Pending to In Progress	The laboratory team is checking the computers.
25 Oct 2026, 07:15 pm	Dr. Rajalakshmi	In Progress	Response: The laboratory computers have been repaired and tested. Students can now use them for practical sessions.
25 Oct 2026, 07:15 pm	Dr. Rajalakshmi	In Progress to Resolved	Maintenance completed and verified.

The assigned faculty member records a response and changes the complaint status. The view includes the response and change history.

3.6 Student Progress Tracking

GRV-0001: Laboratory computers need maintenance

Close details

Infrastructure · Submitted 25 Oct 2026, 07:15 pm

Resolved | Confidential | Student: **Kamal Sundar SK** | Assigned to: **Dr. Rajalakshmi** | Grievance date: 2026-10-25

Two computers in the laboratory freeze during programming sessions. Please arrange maintenance so students can complete their practical work.

Download supporting document: laboratory-note.pdf

Responses

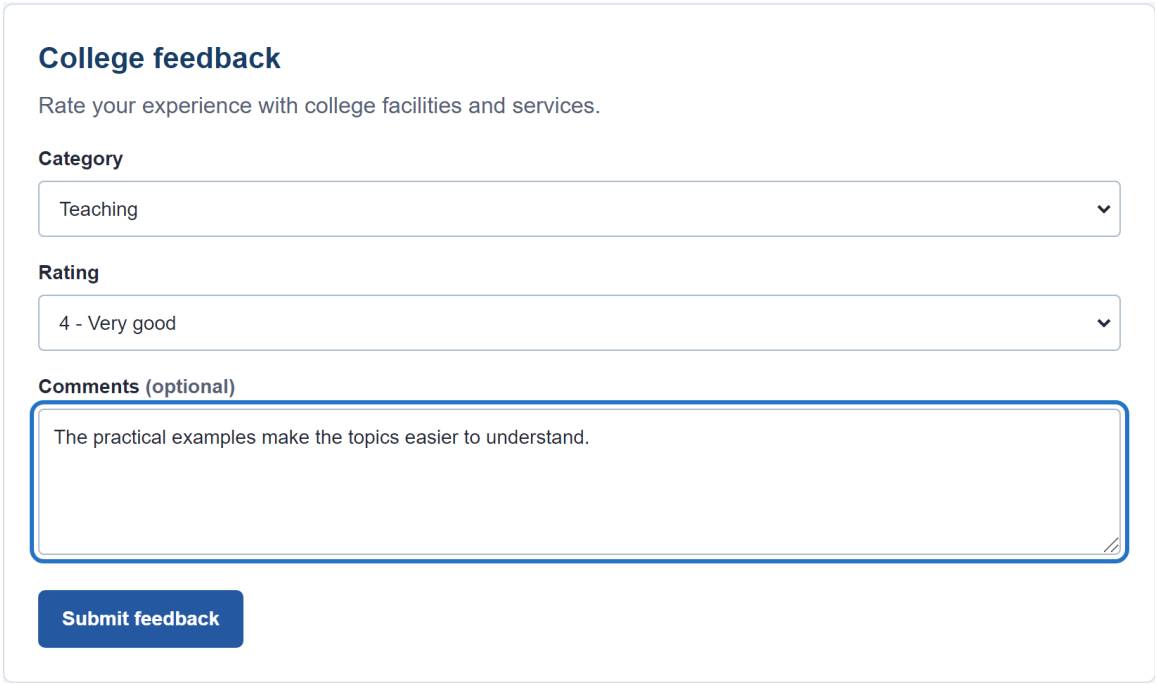
> **Dr. Rajalakshmi** 25 Oct 2026, 07:15 pm
> The laboratory computers have been repaired and tested. Students can now use them for practical sessions.

Grievance history

Date	Updated by	Status change	Remarks
25 Oct 2026, 07:15 pm	Kamal Sundar SK	Pending	Grievance submitted.
25 Oct 2026, 07:15 pm	College Administrator	Pending	Assignment: Unassigned to Dr. Rajalakshmi.
25 Oct 2026, 07:15 pm	Dr. Rajalakshmi	Pending to In Progress	The laboratory team is checking the computers.
25 Oct 2026, 07:15 pm	Dr. Rajalakshmi	In Progress	Response: The laboratory computers have been repaired and tested. Students can now use them for practical sessions.
25 Oct 2026, 07:15 pm	Dr. Rajalakshmi	In Progress to Resolved	Maintenance completed and verified.

The student views the resolved status, faculty response, supporting document and complete complaint history.

3.7 Feedback Form



Students select a college service, give a rating from 1 to 5 and add optional comments.

3.8 Grievance Reports and Feedback Summary

Grievance reports and feedback summary

An overview of complaint resolution and student feedback.

Total grievances	Unresolved	Resolved	Closed
2	1	1	0

By status

Status	Count
Pending	1
In Progress	0
Resolved	1
Closed	0

By category

Category	Count
Academic	0
Infrastructure	1
Examinations	0
Hostel	1
Other	0

Feedback summary

Ratings use a scale of 1 (Poor) to 5 (Excellent). Unresolved includes Pending and In Progress.

Category	Responses	Average rating
Facilities	1	3.0 / 5
Teaching	1	4.0 / 5
Laboratory	1	4.0 / 5
Other Services	1	4.0 / 5

The administrator views complaint totals, unresolved grievances, category and status counts, and average feedback ratings.

4. Conclusion

The Student Grievance and Feedback Management System organizes complaint submission, assignment, response and resolution in one application. React provides the role-based screens, Node.js implements the REST APIs, and MySQL stores the required records. Input validation, password hashing, access restrictions and complaint history support controlled grievance handling. Feedback summaries and grievance statistics help the administrator review college services and pending work.

The interface has been checked using sample API responses, and the database schema has been executed in MySQL. The final live workflow check requires the configured MySQL credentials.